

Algorithms and Data Structures

Problem Sheet 6

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52. Show how to implement a queue using two stacks. Analyze the running time of the queue operations.
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54. Explain why the dynamic-set operation **INSERT** on a singly linked list can be implemented in $\Theta(1)$ time, but the worst-case time for **DELETE** is $\Theta(n)$.
55. Implement a $\Theta(n)$ -time nonrecursive procedure that reverses a singly linked list of n elements. The procedure should use no more than constant storage beyond that needed for the list itself.
56. Write an $O(n)$ recursive procedure that, given an n -node binary tree, prints the key of each node in the tree. Make sure that the key of a node is printed after the keys of their descendants, i.e., the last key printed should be the key of the root.
57. Write an $O(n)$ -time nonrecursive procedure that, given an n -node binary tree, prints out the key of each node in the tree. Use a stack as an auxiliary data structure.
58. Write an $O(n)$ -time nonrecursive procedure that, given an n -node binary tree, prints out the key of each node. Use no more than constant extra space outside of the tree itself and do not modify the tree, even temporarily, during the procedure.
59. Show that any ε -universal family $\mathcal{H}$ of hash functions from a finite set U to a finite set Q has $\varepsilon \geq \frac{1}{|Q|} - \frac{1}{|U|}$.
60. Show that, with double hashing, if m and $h_2(k)$ have greatest common divisor $d > 1$ for some key k , then an unsuccessful search for key k examines $\frac{1}{d}$ th of the hash table before returning to slot $h_1(k)$. Thus, when $d = 1$, so that m and $h_2(k)$ are relatively prime, the search may examine the entire hash table.
61. Consider an open-address hash table with independent uniform permutation hashing and no deletions. Give upper bounds on the expected number of probes in an unsuccessful search and on the expected number of probes in a successful search when the load factor is $\frac{3}{4}$ and when it is $\frac{7}{8}$.